

DATA ENGINEERING OPERATIONS

Automated Pipeline Health Report

Classification: INTERNAL USE ONLY

Date: February 5, 2026

INCIDENT SUMMARY

Field	Details
Incident ID	INC-20260205-ETL-004
Detection Time	2026-02-05 18:05:22 IST
Pipeline Name	PL_RETAIL_DAILY_AGGREGATION
Stage	Step 4: Data Validation & Type Casting
Status	FAILED / CRITICAL
Owner	Data Engineering Team (On-Call)

ERROR DESCRIPTION

The automated ETL pipeline terminated unexpectedly during the **Transformation Phase**. The process encountered a schema validation error while attempting to cast raw input data into the target schema. Specifically, the **gross_margin** column contains non-numeric string values ("NaN" or "TBD") which violates the strict **Float64** requirement for the downstream analytics model.

Impact:

- Daily sales dashboard will not update for the Asia-Pacific region.
- Downstream dependency `Model_Inventory_Forecast_v2` has been paused to prevent data corruption.

TECHNICAL DIAGNOSTICS & LOGS

Host: `worker-node-linux-04`

Execution ID: `exec_88a92_py`

Plaintext

```
[INFO] 18:04:15 - Pipeline initialized. Environment: Production.
[INFO] 18:04:18 - Connection established to Raw_Data_Lake (S3).
[INFO] 18:04:45 - Batch ingestion complete. Rows processed: 45,200.
[INFO] 18:04:50 - Beginning Stage 3: Null Value Imputation.
[INFO] 18:04:55 - Stage 3 complete.
[INFO] 18:04:56 - Beginning Stage 4: Type Casting & Schema Validation.
[WARN] 18:05:10 - Row 14,002: Unexpected format detected in 'transaction_date'.
Auto-corrected.
[ERROR] 18:05:20 - EXCEPTION RAISED IN TASK 'CAST_COLUMNS'
```

Traceback (most recent call last):

```
File "/opt/airflow/dags/scripts/transform_sales.py", line 142, in enforce_schema
    df['gross_margin'] = df['gross_margin'].astype('float64')
File "/usr/local/lib/python3.12/site-packages/pandas/core/dtypes/cast.py", line 120, in astype
    raise ValueError(f"could not convert string to float: '{value}'")
ValueError: could not convert string to float: 'Pending_Audit'
```

```
[FATAL] 18:05:22 - Process terminated with exit code 1.
[INFO] 18:05:23 - Triggering rollback on transaction ID #99821.
[INFO] 18:05:24 - Rollback successful. System state is clean.
```

AUTOMATED REMEDIATION SUGGESTIONS

1. **Immediate Action:** Check the source file `sales_data_2026_02_05.csv` for the string "Pending_Audit" in the `gross_margin` column.
2. **Workaround:** Run the pipeline with the `--quarantine-bad-records` flag to skip erroneous rows and process the remainder of the batch.
3. **Root Cause Investigation:** Verify if the upstream POS (Point of Sale) system API has changed its export format.

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